CUSTOMER BEHAVIOR ANALYSIS

Alfido Tech Data Analytics Internship

Student Details

Name: Kandi Soumya
College: Jyothishmathi Institute of Technology and Science
Internship Role: Data Analytics Intern
Project Title: Customer Behavior Analysis
Tools Used: Python, Pandas, NumPy, Matplotlib, Google Colab
Submission Date: June 2026

Introduction

Customer behavior analysis is essential for understanding purchasing habits, customer preferences, and retention challenges. The objective of this project is to analyze customer transaction data, identify different customer segments, study customer churn patterns, and provide business recommendations to improve customer engagement and overall performance.

Dataset Overview

The dataset used in this project contains customer transaction records from an e-commerce platform. It includes information about customer purchases, demographics, payment methods, product returns, and churn status.

Dataset Details

- Number of Records: 250,000
- Number of Columns: 13
- Unique Customers: 49,673

Key Attributes

- Customer ID
- Purchase Date
- Product Category
- Product Price
- Quantity Purchased
- Total Purchase Amount
- Payment Method
- Customer Age

- Returns
- Gender
- Churn Status

This dataset provides valuable information for understanding customer purchasing behavior and identifying factors that influence customer retention

Data Cleaning and Preparation

Before conducting the analysis, the dataset was cleaned and prepared using Python in Google Colab. The analysis was performed using Python libraries such as Pandas, NumPy, Matplotlib, and Google Colab.

The following steps were performed:

- Examined the dataset structure and verified data types.
- Identified missing values in the Returns column.
- Replaced missing values with zero.
- Converted the Purchase Date column into datetime format.
- Created Month and Year columns for trend analysis.
- Confirmed that no missing values remained in the dataset.

These preprocessing steps ensured that the dataset was accurate, consistent, and suitable for further analysis and visualization.

	0
Customer ID	0
Purchase Date	0
Product Category	0
Product Price	0
Quantity	0
Total Purchase Amount	0
Payment Method	0
Customer Age	0
Returns	0
Customer Name	0
Age	0
Gender	0
Churn	0

dtype: int64

	0
Customer ID	int64
Purchase Date	datetime64[ns]
Product Category	object
Product Price	int64
Quantity	int64
Total Purchase Amount	int64
Payment Method	object
Customer Age	int64
Returns	float64
Customer Name	object
Age	int64
Gender	object
Churn	int64

dtype: object

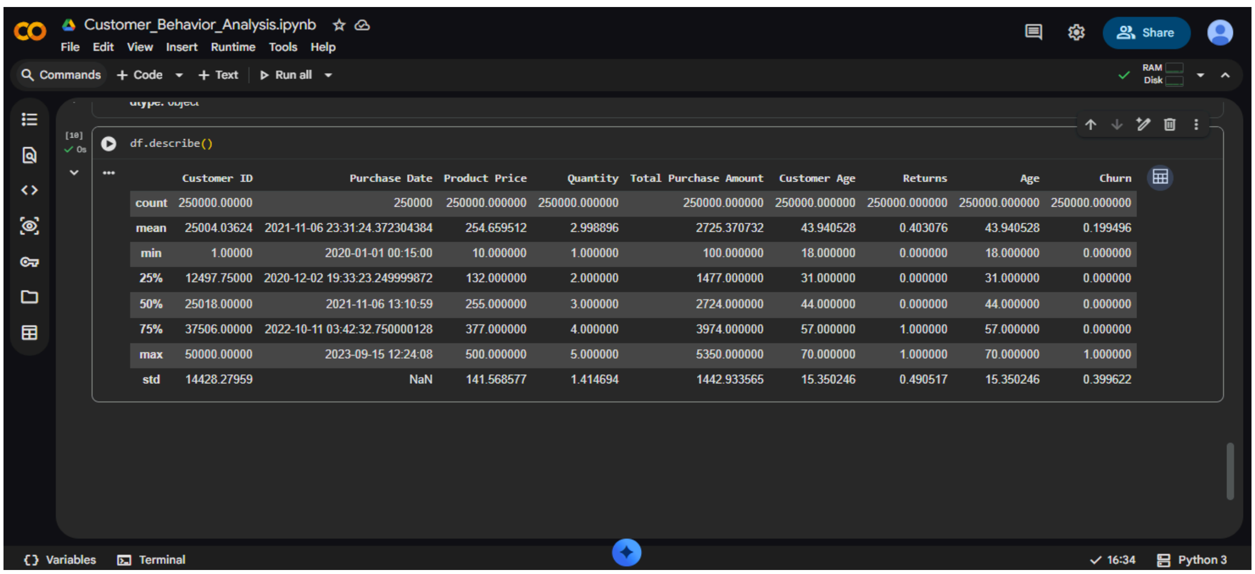
Descriptive Statistics

Basic statistical analysis was performed to understand the distribution of numerical variables within the dataset.

The analysis included:

- Mean
- Median
- Mode
- Data Types
- Summary Statistics

These measures provided insights into customer spending patterns, purchase quantities, age distribution, and other important characteristics.



```
df.describe()
```

	Customer ID	Purchase Date	Product Price	Quantity	Total Purchase Amount	Customer Age	Returns	Age	Churn
count	250000.00000	250000	250000.000000	250000.000000	250000.000000	250000.000000	250000.000000	250000.000000	250000.000000
mean	25004.03624	2021-11-06 23:31:24.372304384	254.659512	2.998896	2725.370732	43.940528	0.403076	43.940528	0.199496
min	1.00000	2020-01-01 00:15:00	10.000000	1.000000	100.000000	18.000000	0.000000	18.000000	0.000000
25%	12497.75000	2020-12-02 19:33:23.249999872	132.000000	2.000000	1477.000000	31.000000	0.000000	31.000000	0.000000
50%	25018.00000	2021-11-06 13:10:59	255.000000	3.000000	2724.000000	44.000000	0.000000	44.000000	0.000000
75%	37506.00000	2022-10-11 03:42:32.750000128	377.000000	4.000000	3974.000000	57.000000	1.000000	57.000000	0.000000
max	50000.00000	2023-09-15 12:24:08	500.000000	5.000000	5350.000000	70.000000	1.000000	70.000000	1.000000
std	14428.27959	NaN	141.568577	1.414694	1442.933565	15.350246	0.490517	15.350246	0.399622

```
Mean
Customer ID              25004.036240
Product Price              254.659512
Quantity                     2.998896
Total Purchase Amount     2725.370732
Customer Age                43.940528
Returns                      0.403076
Age                         43.940528
Churn                        0.199496
dtype: float64

Median
Customer ID              25018.0
Product Price              255.0
Quantity                     3.0
Total Purchase Amount     2724.0
Customer Age                44.0
Returns                      0.0
Age                         44.0
Churn                        0.0
dtype: float64

Mode
Customer ID                            36437
Purchase Date            2022-05-08 12:58:55
Product Category                    Clothing
Product Price                          100.0
Quantity                                 1.0
Total Purchase Amount                 2786.0
Payment Method                   Credit Card
Customer Age                            58.0
Returns                                  0.0
Customer Name                  Michael Smith
Age                                     58.0
Gender                                Female
Churn                                    0.0
Name: 0, dtype: object
```

Customer Segmentation

Customers were segmented based on their total purchase value. This segmentation helps identify customer groups with different spending behaviors and allows businesses to design targeted engagement strategies.

Segment Distribution

Segment	Number of Customers
Low Value	16,560
Medium Value	16,556
High Value	16,557

Customer segmentation was performed using purchase value analysis to classify customers into Low, Medium, and High Value groups.

The results show that customers are distributed almost equally across the three segments. High-value customers are particularly important because they contribute significantly to overall revenue.

	Customer ID	Total Purchase Amount	Quantity
0	1	3491	5
1	2	7988	8
2	3	22587	23
3	4	8715	9
4	5	12524	33

Data Visualizations

1. Customer Segment Distribution

This chart illustrates the distribution of customers across Low Value, Medium Value, and High Value segments.

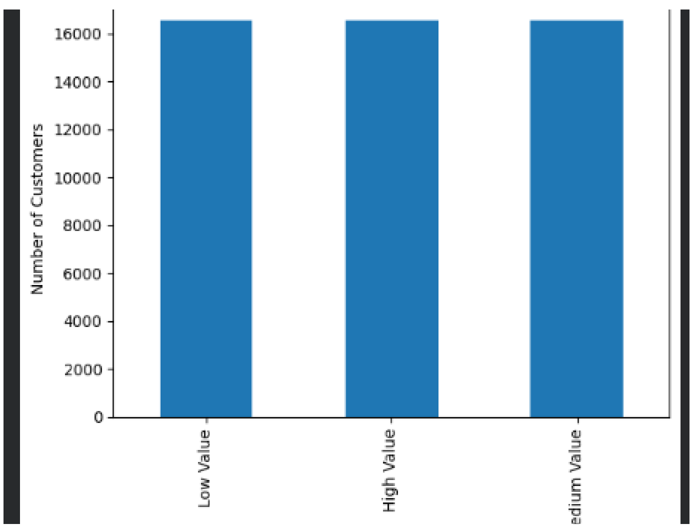


Figure 1: Customer Segment Distribution

2. Product Category Purchases

This visualization highlights customer purchasing activity across different product categories.

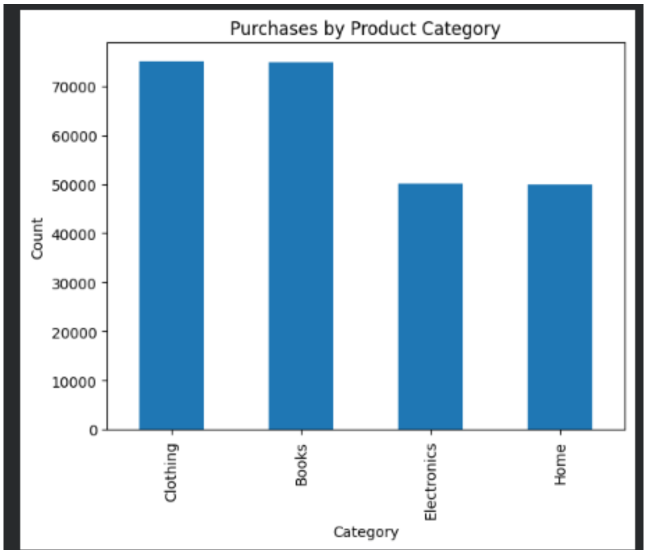


Figure 2: Product Category Purchases

3. Monthly Revenue Trend

The monthly revenue chart was used to analyze purchasing trends and revenue fluctuations over time.

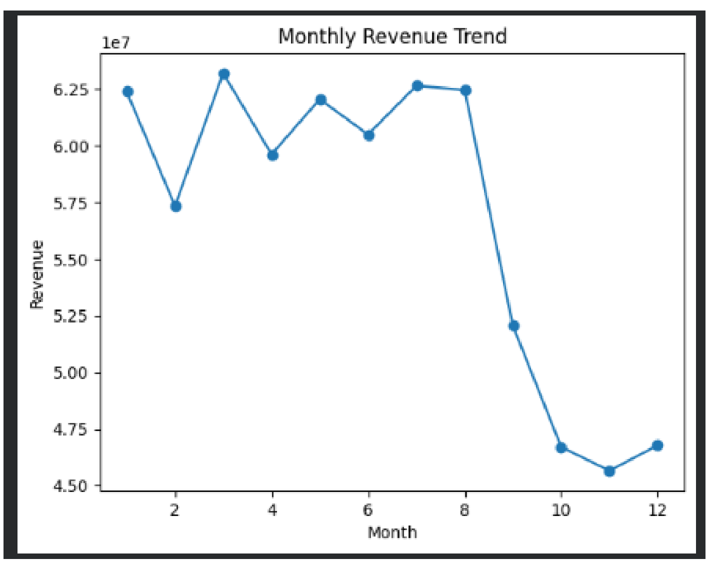


Figure 3: Monthly Revenue Trend

4. Customer Churn Distribution

This chart shows the percentage of customers who remained active and those who churned.

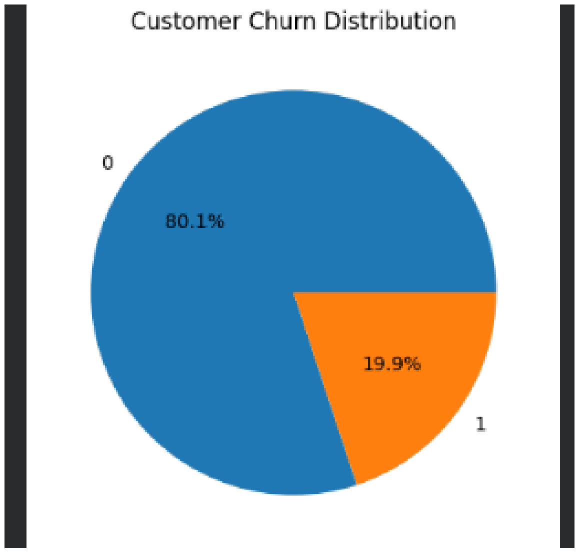


Figure 4: Customer Churn Distribution

5. Payment Method Usage

This visualization displays the most commonly used payment methods among customers.

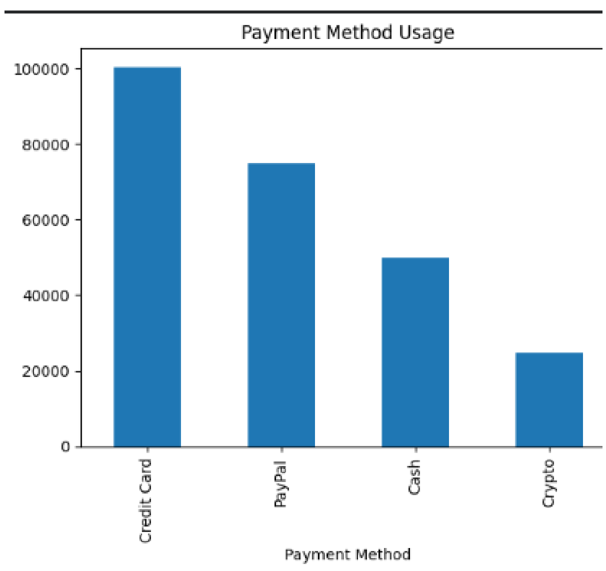


Figure 5: Payment Method Usage

Key Findings

1. The Books category generated the highest revenue, totaling 204,939,601.
2. Clothing was the second-highest revenue-generating category with a total revenue of 204,532,405.

3. Electronics and Home categories generated comparatively lower revenue than Books and Clothing.
4. Approximately 80.05% of customers remained active, indicating a strong customer retention rate.
5. Around 19.95% of customers were identified as churned, highlighting opportunities for retention-focused strategies.
6. High-value customers represent a crucial customer segment and contribute significantly to overall business revenue.
7. Purchasing behavior varies across product categories, creating opportunities for targeted marketing campaigns.

Recommendations

Based on the analysis, the following recommendations are suggested:

1. Introduce loyalty and rewards programs for high-value customers to encourage repeat purchases.
2. Provide personalized product recommendations based on customer purchase history and preferences.
3. Develop targeted retention campaigns for customers who are at a higher risk of churning.
4. Increase promotional efforts for high-performing product categories such as Books and Clothing.
5. Improve the customer experience by maintaining convenient payment options and a smooth checkout process.

Conclusion

The customer behavior analysis provided valuable insights into purchasing trends, customer segmentation, and churn patterns. The findings indicate that customer retention and personalized engagement strategies can significantly improve business performance. By focusing on high-value customers, reducing churn, and strengthening marketing efforts in top-performing product categories, businesses can enhance customer satisfaction and support long-term growth.

EXECUTIVE SUMMARY

This project analyzed customer transaction data to understand purchasing behavior, customer segmentation, and churn patterns. A total of 250,000 transaction records were examined and customers were segmented into Low Value, Medium Value, and High Value groups based on their purchase amounts. The analysis showed that Books and Clothing were the highest revenue-generating product categories. Approximately 20% of customers were identified as churned, indicating opportunities for retention strategies. Based on the findings, recommendations such as loyalty programs, personalized marketing campaigns, and targeted customer engagement initiatives were suggested to improve customer satisfaction and business performance.